# Snort 3 Intrusion Detection System (IDS) on Kali Linux

**Name:** Chetna Solanki

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## 1. Introduction

Snort 3 is an open-source Network Intrusion Detection System (NIDS) developed by Cisco. It monitors network traffic in real time and analyzes packets using predefined and custom rules. Whenever suspicious or malicious traffic matches a rule, Snort generates an alert to help security analysts investigate the activity.

In this practical, Snort 3 was installed on Kali Linux, configured successfully, and tested in a virtual lab environment. Network traffic was generated from a Windows 11 virtual machine to observe how Snort monitors and analyzes packets.

## 2. Objective

- Install Snort 3 on Kali Linux.
- Verify the Snort installation.
- Configure Snort for network monitoring.
- Create and verify local detection rules.
- Run Snort in IDS mode.
- Generate traffic from a Windows 11 virtual machine.
- Monitor and analyze detected network activity.

## 3. Lab Setup

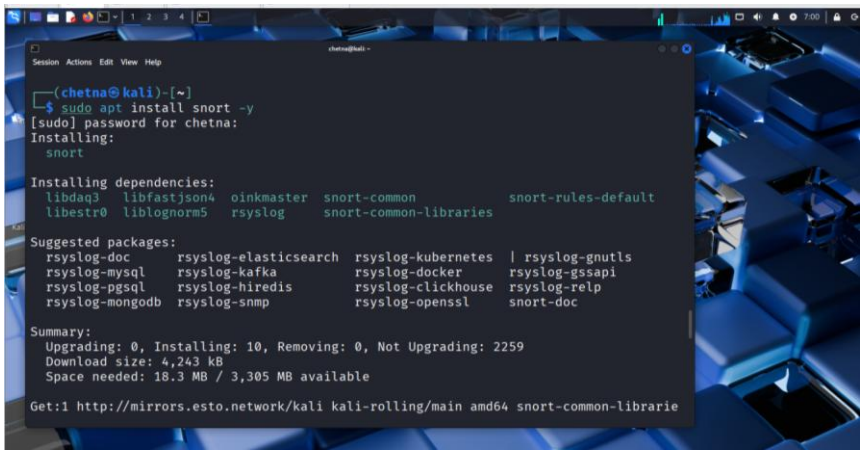
| Component | Details |
|---|---|
| Operating System | Kali Linux |
| IDS Tool | Snort 3 |
| Attacker Machine | Windows 11 VM |
| Target Machine | Kali Linux |

| Component | Details |
| --- | --- |
| Network Interface | eth0 |
| Attack Tool | Ping (ICMP) |

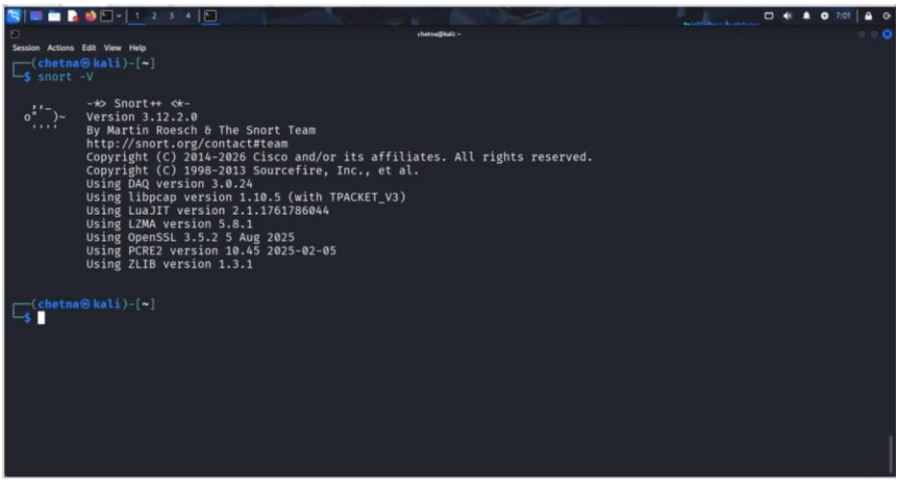
## 4. Installing Snort

*Update packages : sudo apt update*

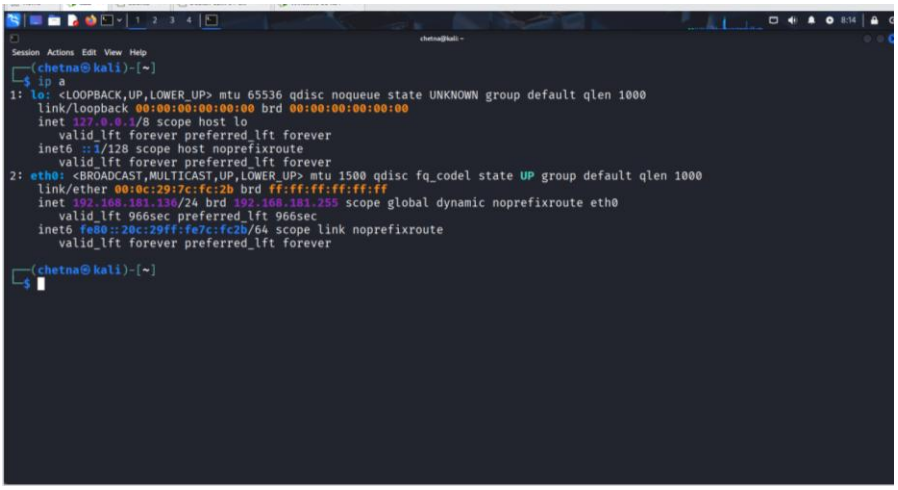
*Install Snort : sudo apt install snort -y*




*Check version : snort -V*



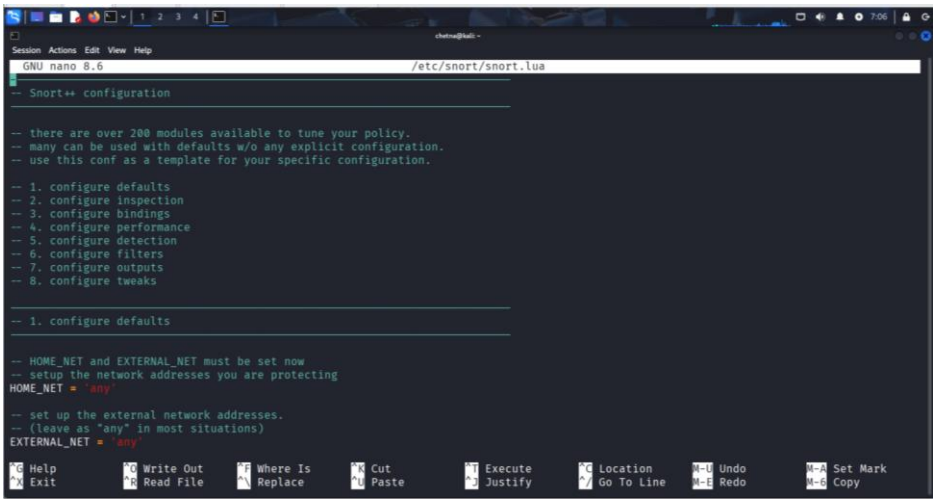

## 5. Network Interface Verification



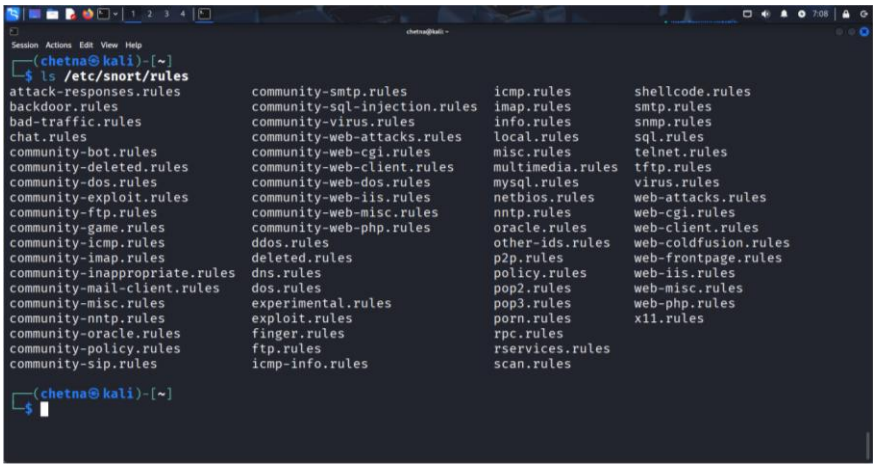

# 6. Snort Configuration

## Open the main configuration file.

*sudo nano /etc/snort/snort.conf*

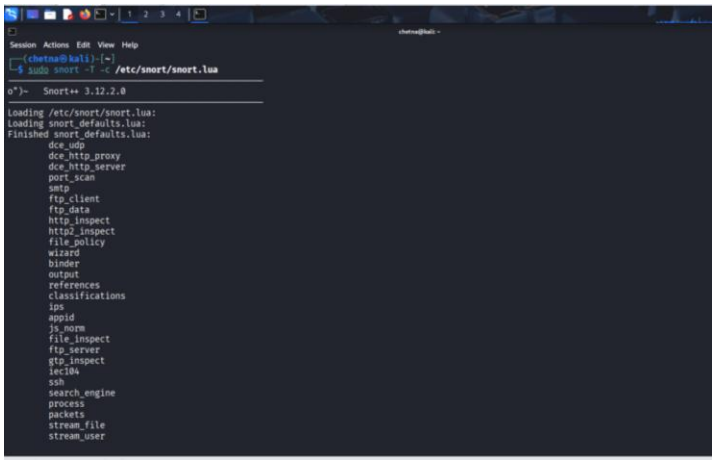


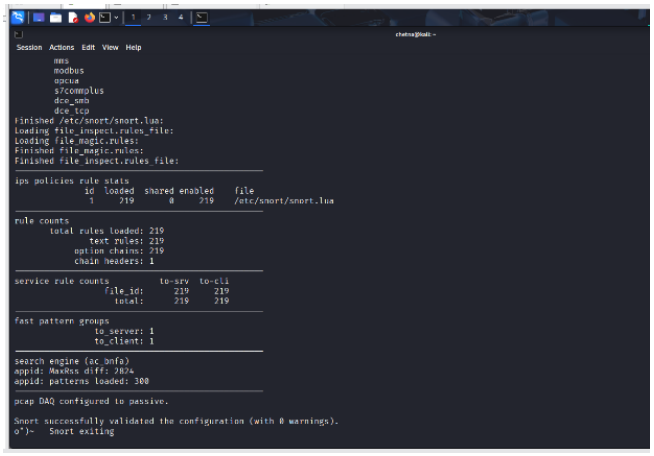

The snort.lua file contains the primary configuration settings for Snort 3, including inspection modules, detection settings, logging options, and rule loading.



Validate configuration.

*sudo snort -T -c /etc/snort/snort.conf*

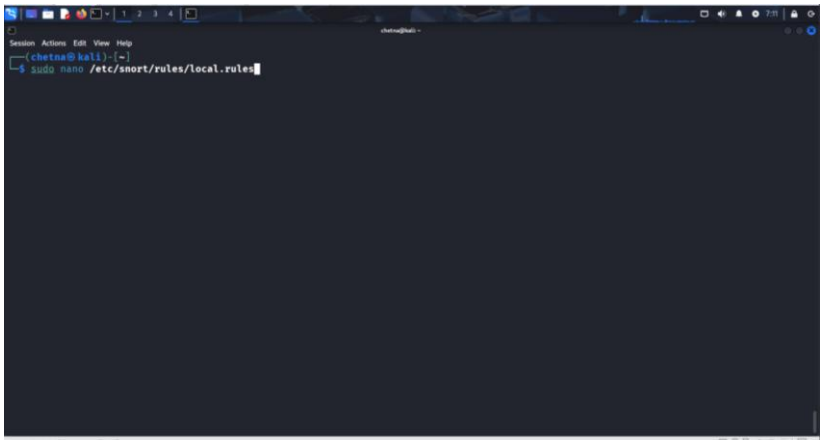



# 7. Creating a Local Detection Rule

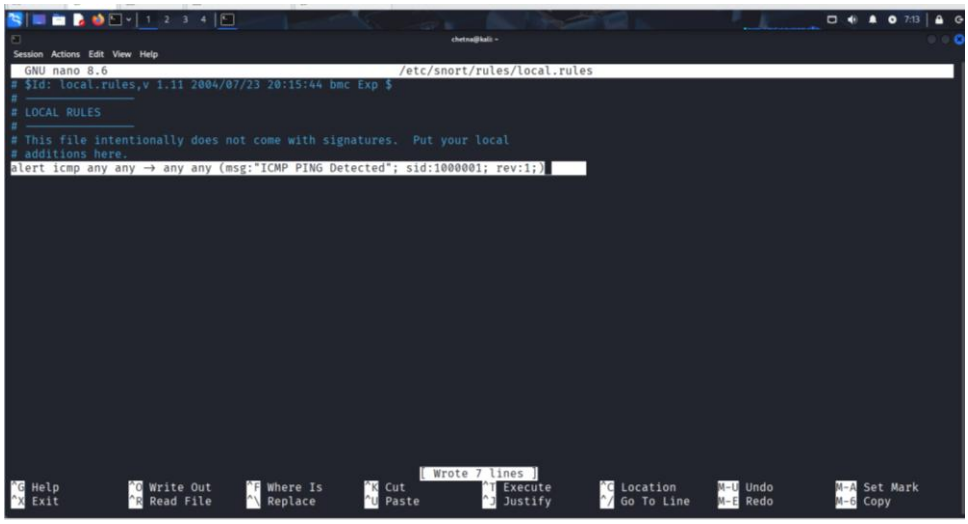
A custom rule was created to detect ICMP (Ping) traffic during testing.

Open the local rules file:




Add rule.

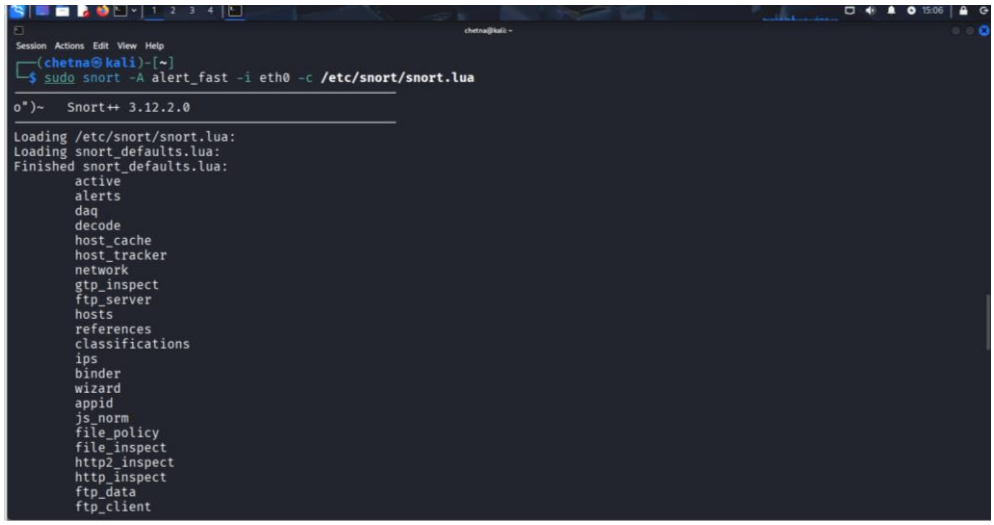
alert icmp any any -> any any (msg:"ICMP PING Detected"; sid:1000001; rev:1;)

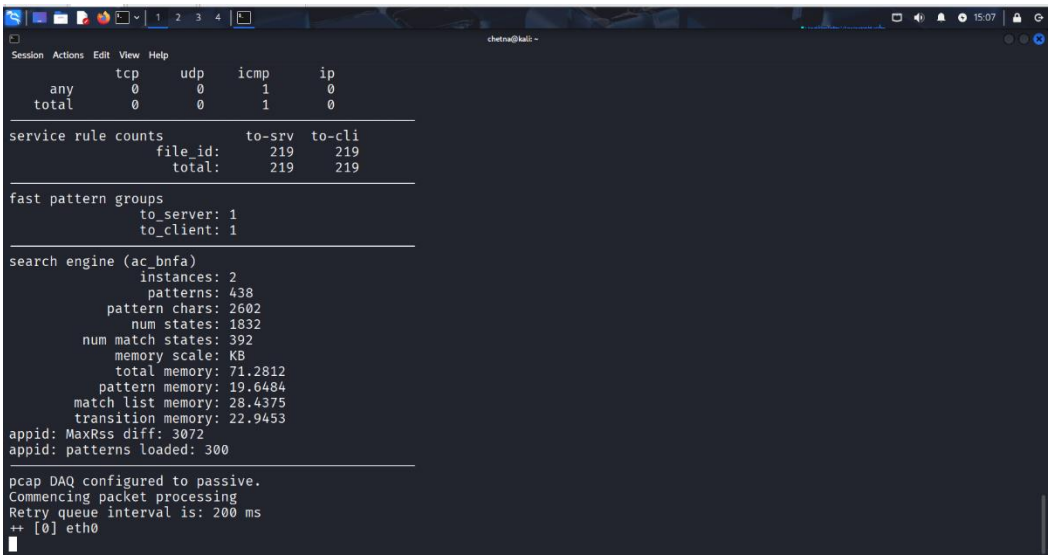



This rule instructs Snort to generate an alert whenever an ICMP packet is detected.

## 8. Running Snort

After verifying the configuration, Snort was started in IDS mode to monitor live network traffic.



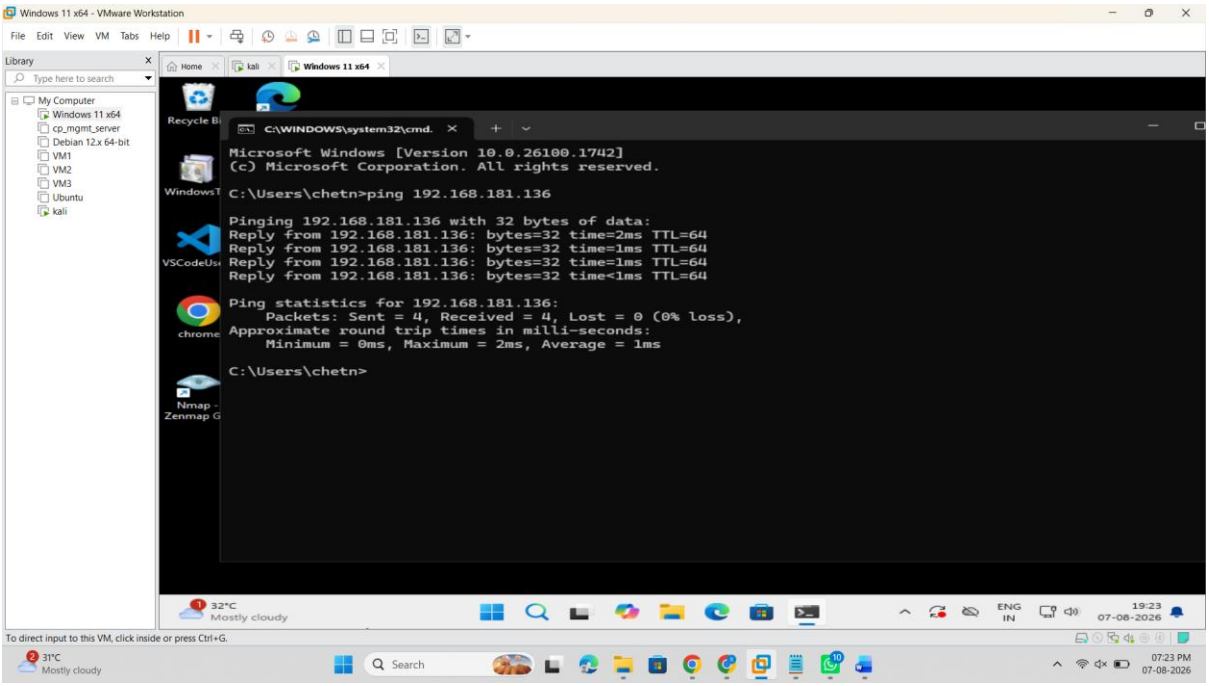



**9: Simulating ICMP Traffic**

To verify the custom detection rule, ICMP (Ping) traffic was generated from the Windows 11 virtual machine to the Kali Linux system.

Command:

ping 192.168.181.136




**Learning:**

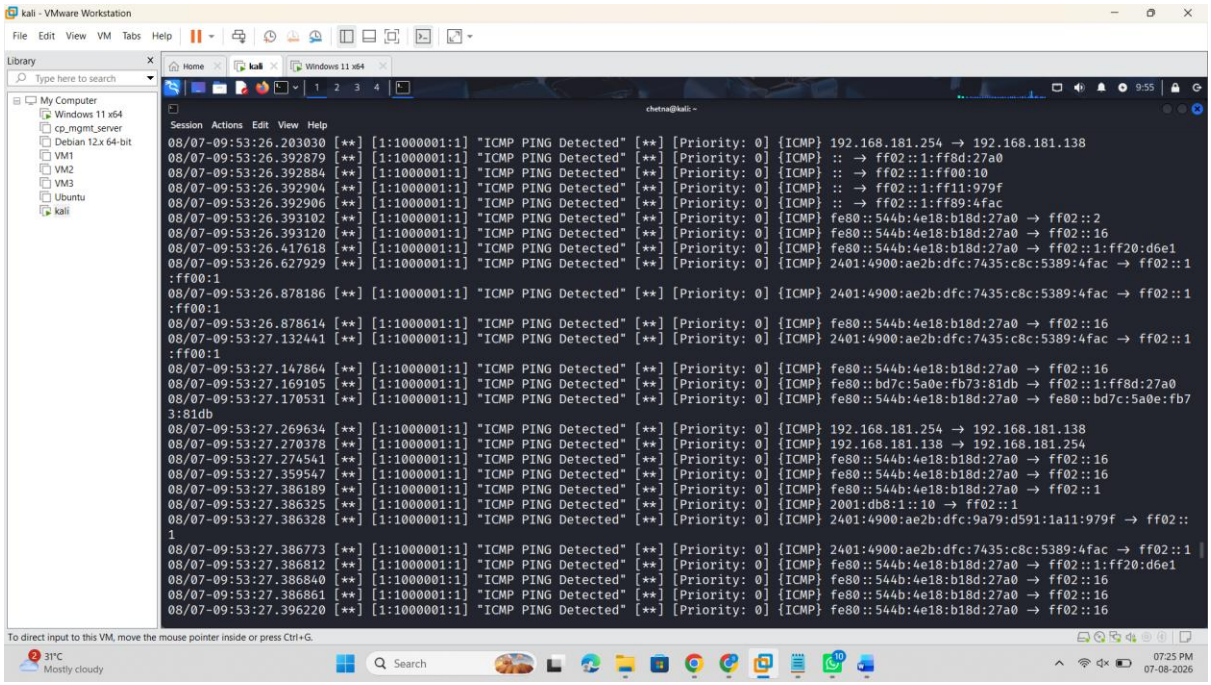
**This test demonstrates how Snort detects and analyzes ICMP traffic in real time.**

**10 :Snort Alert Detection**

While the Ping test was running, Snort inspected the incoming ICMP packets and matched them with the configured detection rule.

The generated alert included:

- Alert Message
- Source IP Address
- Destination IP Address
- Protocol Type (ICMP)
- Timestamp




**Result:**
Snort successfully detected the ICMP Ping traffic and generated an alert based on the configured rule.

## 11. Alert Analys

The generated alert confirms that Snort successfully inspected the network traffic and identified the simulated reconnaissance activity.

## Observed Result

| Parameter | Value |
|---|---|
| Attacker | Windows 11 VM |

| **Parameter** | **Value** |
| --- | --- |
| Target | Kali Linux |
| Tool | Snort 3 |
| Monitoring Status | Successful |
| Attack tool | Ping (ICMP) |

The alert information helps security analysts understand the source of the activity and supports further investigation.